

Power Electronics and Integrated Circuit Control –

Homework 1

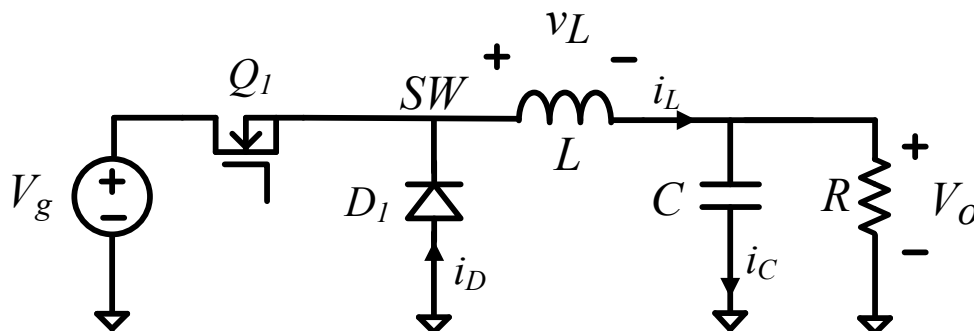
Date: 2025/3/5

Due date: 3/19

1. (45%) Lecture PEIC-2 p.47~49 group discussion related questions (total 3 questions)

- (a) V_o Waveform of boost converter if C_o has ESR and ESL? *→ spike at edge*
(b) Give three reasons why **boost** converter $V_o \geq V_{in}$.
(c) Derivation of DCM Boost Converter voltage gain as a function of duty cycle and load resistance

2. A Buck converter in steady-state operation



- (a) (25%) Critical resistance is defined as the value of load resistance R , which achieves operation at the CCM/DCM boundary. Assume $V_g = 10$ V, $V_o = 0.5$ V, Diode D_1 has a forward voltage of 0.5 V, $L = 10$ μ H, $C = 100$ μ F, and Switching frequency $f_s = 2$ MHz, where R is 10% of critical resistance.

Derive critical resistance value, peak and valley values of V_o , i_L , i_C , SW , and duty cycle D .

- (b) (30%) Simulate the converter in (a) to get the steady-state time waveform as p. 42 in week 2 lecture slide. Paste the simulated circuit picture and the simulated waveforms. Mark the important values such as peak and valley currents/voltages and D to prove your calculation in (a). Discuss its difference with your calculation.